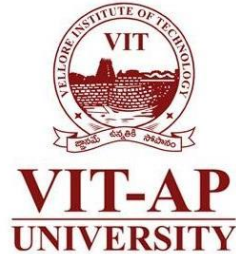


MAT1001 Calculus for Engineers

Green's Theorem



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Flow and Circulation

DEFINITIONS If $\mathbf{r}(t)$ parametrizes a smooth curve C in the domain of a continuous velocity field $\mathbf{F}$, the **flow** along the curve from $A = \mathbf{r}(a)$ to $B = \mathbf{r}(b)$ is

$$\text{Flow} = \int_C \mathbf{F} \cdot \mathbf{T} \, ds. \quad (5)$$

The integral in this case is called a **flow integral**. If the curve starts and ends at the same point, so that $A = B$, the flow is called the **circulation** around the curve.

The direction we travel along C matters. If we reverse the direction, then $\mathbf{T}$ is replaced by $-\mathbf{T}$ and the sign of the integral changes.

EXAMPLE A fluid's velocity field is $\mathbf{F} = x\mathbf{i} + z\mathbf{j} + y\mathbf{k}$. Find the flow along the helix $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$, $0 \leq t \leq \pi/2$.

Solution $\mathbf{F} = x\mathbf{i} + z\mathbf{j} + y\mathbf{k} = (\cos t)\mathbf{i} + t\mathbf{j} + (\sin t)\mathbf{k}$ $\frac{d\mathbf{r}}{dt} = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j} + \mathbf{k}$

$$\begin{aligned} \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} &= (\cos t)(-\sin t) + (t)(\cos t) + (\sin t)(1) \\ &= -\sin t \cos t + t \cos t + \sin t \end{aligned}$$

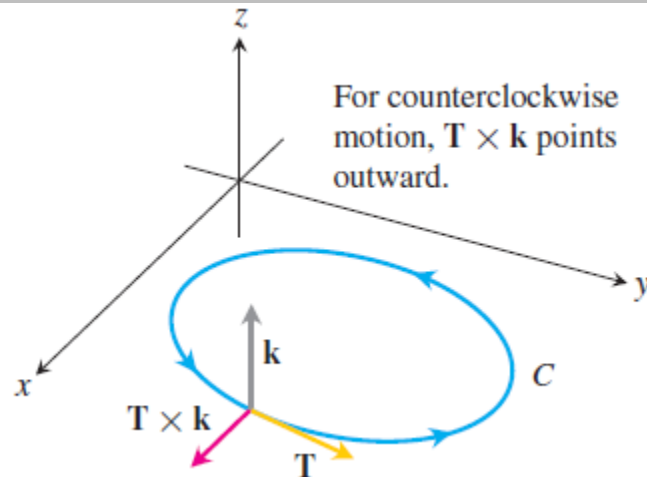
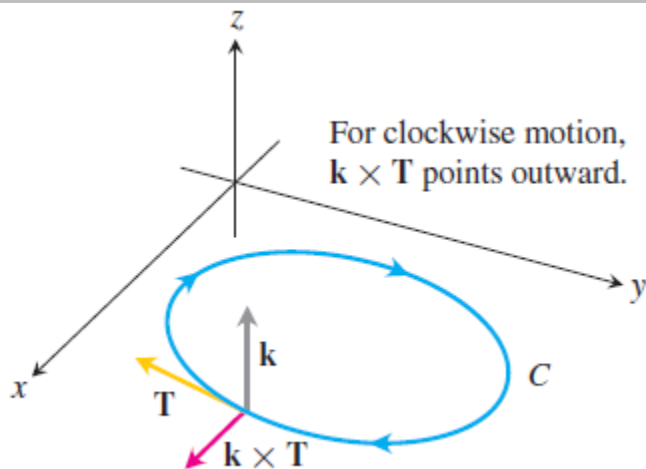
$$\text{Flow} = \int_{t=a}^{t=b} \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} \, dt = \frac{\pi}{2} - \frac{1}{2}$$

DEFINITION If C is a smooth simple closed curve in the domain of a continuous vector field $\mathbf{F} = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$ in the plane, and if $\mathbf{n}$ is the outward-pointing unit normal vector on C , the **flux** of $\mathbf{F}$ across C is

$$\text{Flux of } \mathbf{F} \text{ across } C = \int_C \mathbf{F} \cdot \mathbf{n} \, ds. \quad (6)$$

Notice the difference between flux and circulation. The flux of $\mathbf{F}$ across C is the line integral with respect to arc length of $\mathbf{F} \cdot \mathbf{n}$, the scalar component of $\mathbf{F}$ in the direction of the outward normal. The circulation of $\mathbf{F}$ around C is the line integral with respect to arc length of $\mathbf{F} \cdot \mathbf{T}$, the scalar component of $\mathbf{F}$ in the direction of the unit tangent vector. Flux is the integral of the normal component of $\mathbf{F}$; circulation is the integral of the tangential component of $\mathbf{F}$.

Flux - Evaluation



we are about to derive for computing $\mathbf{n}$ and evaluating the integral assume counterclockwise motion.

$$\mathbf{n} = \mathbf{T} \times \mathbf{k} = \left(\frac{dx}{ds} \mathbf{i} + \frac{dy}{ds} \mathbf{j} \right) \times \mathbf{k} = \frac{dy}{ds} \mathbf{i} - \frac{dx}{ds} \mathbf{j}.$$

$$\mathbf{F} = M(x, y) \mathbf{i} + N(x, y) \mathbf{j}$$

$$\mathbf{F} \cdot \mathbf{n} = M(x, y) \frac{dy}{ds} - N(x, y) \frac{dx}{ds}.$$

$$\int_C \mathbf{F} \cdot \mathbf{n} \, ds = \int_C \left(M \frac{dy}{ds} - N \frac{dx}{ds} \right) ds = \oint_C M \, dy - N \, dx.$$

Flux - Evaluation

Calculating Flux Across a Smooth Closed Plane Curve

$$(\text{Flux of } \mathbf{F} = M\mathbf{i} + N\mathbf{j} \text{ across } C) = \oint_C M dy - N dx \quad (7)$$

The integral can be evaluated from any smooth parametrization $x = g(t)$, $y = h(t)$, $a \leq t \leq b$, that traces C counterclockwise exactly once.

EXAMPLE Find the flux of $\mathbf{F} = (x - y)\mathbf{i} + x\mathbf{j}$ across the circle $x^2 + y^2 = 1$ in the xy -plane.

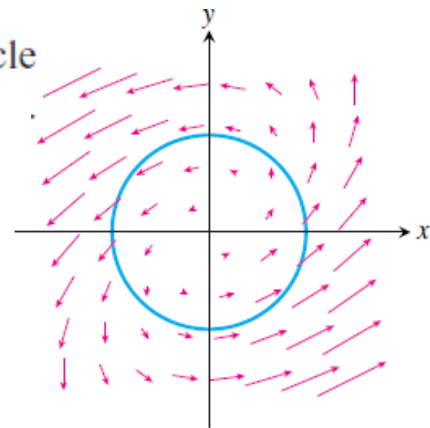
Solution The parametrization $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}$, $0 \leq t \leq 2\pi$, traces the circle counterclockwise exactly once.

$$M = x - y = \cos t - \sin t, \quad dy = d(\sin t) = \cos t dt$$

$$N = x = \cos t, \quad dx = d(\cos t) = -\sin t dt$$

$$\text{Flux} = \int_C M dy - N dx = \int_0^{2\pi} (\cos^2 t - \sin t \cos t + \cos t \sin t) dt = \pi.$$

The flux of $\mathbf{F}$ across the circle is π . Since the answer is positive, the net flow across the curve is outward. A net inward flow would have given a negative flux. ■



Divergence – Flux Density

DEFINITION

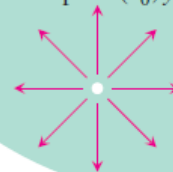
The **divergence (flux density)** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is

$$\operatorname{div} \mathbf{F} = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}. \quad (1)$$

A gas is compressible, unlike a liquid, and the divergence of its velocity field measures to what extent it is expanding or compressing at each point. Intuitively, if a gas is expanding at the point (x_0, y_0) , the lines of flow would diverge there (hence the name) and, since the gas would be flowing out of a small rectangle about (x_0, y_0) , the divergence of $\mathbf{F}$ at (x_0, y_0) would be positive. If the gas were compressing instead of expanding, the divergence would be negative

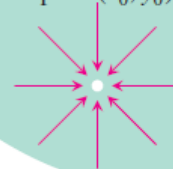
Source: $\operatorname{div} \mathbf{F}(x_0, y_0) > 0$

A gas expanding at the point (x_0, y_0) .



Sink: $\operatorname{div} \mathbf{F}(x_0, y_0) < 0$

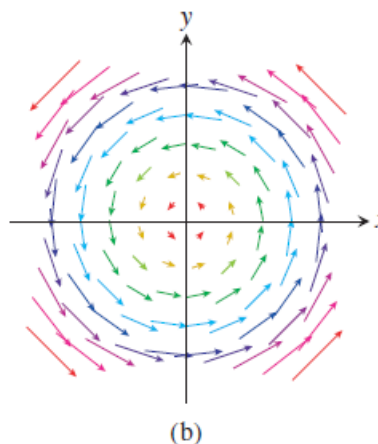
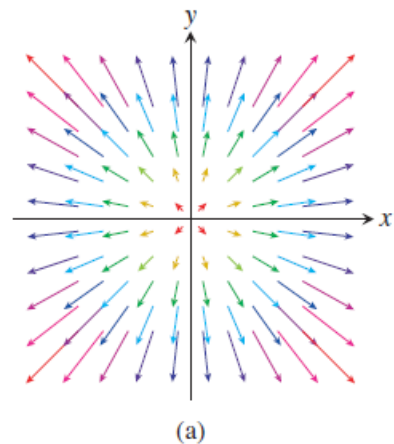
A gas compressing at the point (x_0, y_0) .



Example

EXAMPLE

The following vector fields represent the velocity of a gas flowing in the xy -plane. Find the divergence of each vector field and interpret its physical meaning.



(a) *Uniform expansion or compression:* $\mathbf{F}(x, y) = cx\mathbf{i} + cy\mathbf{j}$

(b) *Uniform rotation:* $\mathbf{F}(x, y) = -cy\mathbf{i} + cx\mathbf{j}$

Solution

(a) $\text{div } \mathbf{F} = \frac{\partial}{\partial x}(cx) + \frac{\partial}{\partial y}(cy) = 2c$: If $c > 0$, the gas is undergoing uniform expansion; if $c < 0$, it is undergoing uniform compression.

(b) $\text{div } \mathbf{F} = \frac{\partial}{\partial x}(-cy) + \frac{\partial}{\partial y}(cx) = 0$: The gas is neither expanding nor compressing.

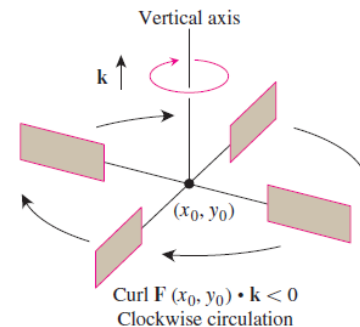
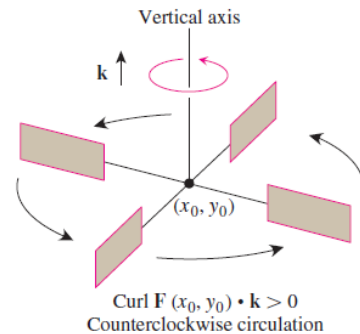
Curl and Circulation Density

DEFINITION The **circulation density** of a vector field $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ at the point (x, y) is the scalar expression

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \quad (2)$$

This expression is also called the **k-component of the curl**, denoted by $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$.

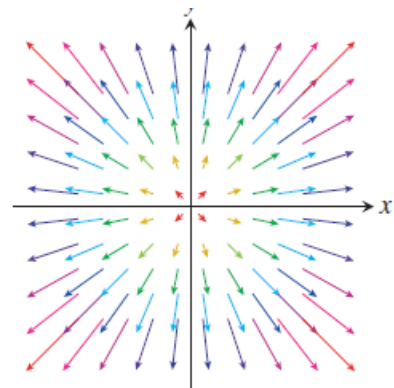
If water is moving about a region in the xy -plane in a thin layer, then the **k**-component of the curl at a point (x_0, y_0) gives a way to measure how fast and in what direction a small paddle wheel spins if it is put into the water at (x_0, y_0) with its axis perpendicular to the plane, parallel to $\mathbf{k}$



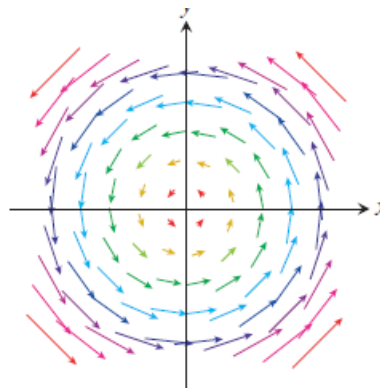
Example

EXAMPLE

Find the circulation density, and interpret what it means, for each vector field in Example



(a)



(b)

(a) *Uniform expansion or compression:* $\mathbf{F}(x, y) = cx\mathbf{i} + cy\mathbf{j}$

(b) *Uniform rotation:* $\mathbf{F}(x, y) = -cy\mathbf{i} + cx\mathbf{j}$

Solution

(a) *Uniform expansion:* $(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial x}(cy) - \frac{\partial}{\partial y}(cx) = 0$. The gas is not circulating at very small scales.

(b) *Rotation:* $(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial}{\partial x}(cx) - \frac{\partial}{\partial y}(-cy) = 2c$. The constant circulation density indicates rotation at every point. If $c > 0$, the rotation is counterclockwise; if $c < 0$, the rotation is clockwise.

THEOREM (Green's Theorem) *Let R be a simply connected plane region whose boundary is a simple, closed, piecewise smooth curve C oriented counterclockwise. If $f(x, y)$ and $g(x, y)$ are continuous and have continuous first partial derivatives on some open set containing R , then*

$$\int_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA \quad (1)$$

Example

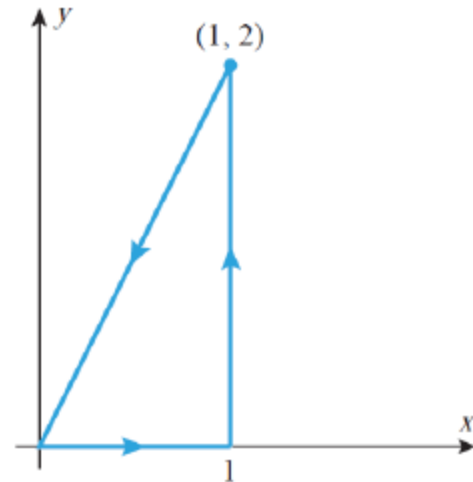
► **Example** Use Green's Theorem to evaluate

$$\int_C x^2 y dx + x dy$$

along the triangular path shown in Figure

Solution. Since $f(x, y) = x^2 y$ and $g(x, y) = x$, it follows from (1) that

$$\begin{aligned} \int_C x^2 y dx + x dy &= \iint_R \left[\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(x^2 y) \right] dA = \int_0^1 \int_0^{2x} (1 - x^2) dy dx \\ &= \int_0^1 (2x - 2x^3) dx = \left[x^2 - \frac{x^4}{2} \right]_0^1 = \frac{1}{2} \end{aligned}$$



Notation for Line Integral

It is common practice to denote a line integral around a simple closed curve by an integral sign with a superimposed circle. With this notation Formula (1) would be written as

$$\oint_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

Sometimes a direction arrow is added to the circle to indicate whether the integration is clockwise or counterclockwise. Thus, if we wanted to emphasize the counterclockwise direction of integration required by Theorem , we could express (1) as

$$\oint_C f(x, y) dx + g(x, y) dy = \iint_R \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dA$$

Finding Work Using Green's Theorem

► **Example** Find the work done by the force field

$$\mathbf{F}(x, y) = (e^x - y^3)\mathbf{i} + (\cos y + x^3)\mathbf{j}$$

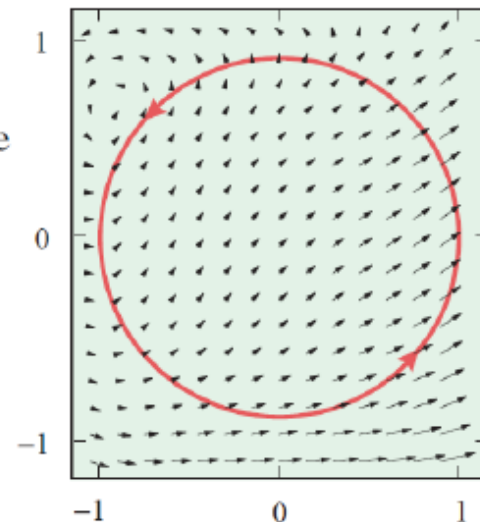
on a particle that travels once around the unit circle $x^2 + y^2 = 1$ in the counterclockwise direction

Solution. The work W performed by the field is

$$\begin{aligned} W &= \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (e^x - y^3) dx + (\cos y + x^3) dy \\ &= \iint_R \left[\frac{\partial}{\partial x} (\cos y + x^3) - \frac{\partial}{\partial y} (e^x - y^3) \right] dA \\ &= \iint_R (3x^2 + 3y^2) dA = 3 \iint_R (x^2 + y^2) dA \\ &= 3 \int_0^{2\pi} \int_0^1 (r^2) r dr d\theta = \frac{3}{4} \int_0^{2\pi} d\theta = \frac{3\pi}{2} \quad \blacktriangleleft \end{aligned}$$

We converted to polar coordinates.

Green's Theorem



Finding Area of a Region Using Green's Theorem

Green's Theorem leads to some useful new formulas for the area A of a region R that satisfies the conditions of the theorem. The following three formulas that express the area A of a region R in terms of line integrals around the boundary:

$$A = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C -y \, dx + x \, dy$$

► **Example** Use a line integral to find the area enclosed by the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Solution. The ellipse, with counterclockwise orientation, can be represented parametrically by

$$x = a \cos t, \quad y = b \sin t \quad (0 \leq t \leq 2\pi)$$

If we denote this curve by C , then from the third formula in (6) the area A enclosed by the ellipse is

$$\begin{aligned} A &= \frac{1}{2} \oint_C -y \, dx + x \, dy = \frac{1}{2} \int_0^{2\pi} [(-b \sin t)(-a \sin t) + (a \cos t)(b \cos t)] \, dt \\ &= \frac{1}{2} ab \int_0^{2\pi} (\sin^2 t + \cos^2 t) \, dt = \frac{1}{2} ab \int_0^{2\pi} dt = \pi ab \end{aligned}$$

THEOREM —Green's Theorem (Flux-Divergence or Normal Form) Let C be a piecewise smooth, simple closed curve enclosing a region R in the plane. Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R . Then the outward flux of $\mathbf{F}$ across C equals the double integral of $\text{div } \mathbf{F}$ over the region R enclosed by C .

$$\oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \oint_C M \, dy - N \, dx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy$$

Outward flux Divergence integral

THEOREM —Green's Theorem (Circulation-Curl or Tangential Form) Let C be a piecewise smooth, simple closed curve enclosing a region R in the plane. Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R . Then the counterclockwise circulation of $\mathbf{F}$ around C equals the double integral of $(\text{curl } \mathbf{F}) \cdot \mathbf{k}$ over R .

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \oint_C M \, dx + N \, dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx \, dy$$

Counterclockwise circulation

Curl integral

Example

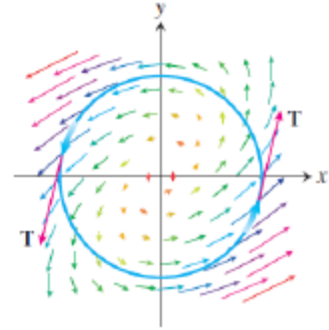
► **Example** Verify both forms of Green's Theorem for the vector field $\mathbf{F}(x, y) = (x - y)\mathbf{i} + x\mathbf{j}$ and the region R bounded by the unit circle $C: \mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j}, \quad 0 \leq t \leq 2\pi$.

Solution. Evaluating $\mathbf{F}(\mathbf{r}(t))$ and differentiating components, we have

$$M = \cos t - \sin t, \quad dx = d(\cos t) = -\sin t \, dt,$$

$$N = \cos t, \quad dy = d(\sin t) = \cos t \, dt,$$

$$\frac{\partial M}{\partial x} = 1, \quad \frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1, \quad \frac{\partial N}{\partial y} = 0.$$



With the Normal Form

$$\oint_C M \, dy - N \, dx = \int_{t=0}^{t=2\pi} (\cos t - \sin t)(\cos t \, dt) - (\cos t)(-\sin t \, dt) = \int_0^{2\pi} \cos^2 t \, dt = \pi$$

$$\iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx \, dy = \iint_R (1 + 0) \, dx \, dy = \iint_R dx \, dy = \text{area inside the unit circle} = \pi.$$

With the Tangential Form

$$\oint_C M dx + N dy = \int_{t=0}^{t=2\pi} (\cos t - \sin t)(-\sin t dt) + (\cos t)(\cos t dt)$$

$$= \int_0^{2\pi} (-\sin t \cos t + 1) dt = 2\pi$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \iint_R (1 - (-1)) dx dy = 2 \iint_R dx dy = 2\pi.$$

Example

► **Example** Evaluate the line integral

$$\oint_C xy \, dy - y^2 \, dx,$$

where C is the square cut from the first quadrant by the lines $x = 1$ and $y = 1$.

Solution. We can use either form of Green's Theorem to change the line integral into a double integral over the square.

1. *With the Normal Form* : Taking $M = xy$, $N = y^2$, and C and R as the square's boundary and interior gives

$$\oint_C xy \, dy - y^2 \, dx = \iint_R (y + 2y) \, dx \, dy = \int_0^1 \int_0^1 3y \, dx \, dy = \int_0^1 \left[3xy \right]_{x=0}^{x=1} dy = \int_0^1 3y \, dy = \left[\frac{3}{2} y^2 \right]_0^1 = \frac{3}{2}.$$

2. *With the Tangential Form* : Taking $M = -y^2$ and $N = xy$ gives the same result:

$$\oint_C -y^2 \, dx + xy \, dy = \iint_R (y - (-2y)) \, dx \, dy = \frac{3}{2}.$$

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Thank You...

A large, faded version of the VIT-AP University logo is centered in the background of the slide. It consists of the text "VIT-AP" in a large, serif font, with "UNIVERSITY" in a smaller, serif font below it, separated by a horizontal line.

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